

PRIYA ALAGARSAMY, Ph.D.

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SUMMARY

Data science leader with a Ph.D. in Finance and 10+ years' experience, including three years leading and mentoring data science teams. Delivered \$18 million in measurable enterprise value across pricing, monetization, and marketing through experimentation, causal inference, and production-grade AI/ML models. Proven track record partnering with Product, Marketing, Engineering, and Finance to shape roadmaps and translate complex quantitative signals into commercial decisions. Deep fluency in experimentation design and attribution (A/B testing, causal inference), pricing and monetization analytics (price elasticity & optimization via LP, MIP, non-linear methods), customer segmentation, and revenue/lifecycle forecasting (regression, logistic, XGBoost, time-series). Built scalable, self-serve systems and AI-enabled insight generation (cloud-native pipelines, GenAI/multi-agent frameworks) that increase team leverage. Strong coding skills in SQL, Python, GCP, PySpark, TensorFlow.

PROFESSIONAL EXPERIENCE

Scotiabank

Toronto, Canada

Senior Manager/Senior Data Scientist, Pricing/Advanced Analytics Modeling

July 2023 – Present

- Defined the measurement and experimentation strategy for GIC pricing optimization, owning core KPIs and using A/B testing and causal inference to quantify incremental ROI on pricing investments.
- Led customer price-sensitivity segmentation (tree-based panel model) to identify high-value, price-sensitive clients, partnering with Pricing and Marketing leadership to deliver \$4.6 million in incremental value.
- Designed geographic segmentation using macroeconomics, habits, advisor behaviour features for Canadian districts to reduce customer drop-off in pricing journeys.
- Partnered with Pricing leadership to redesign the pricing optimization model around the re-developed sensitivity segments, driving a 6% improvement in margin with no volume change — a direct commercial outcome of the experimentation work.
- Led development of a production-grade propensity model to identify retention risk, enabling targeted marketing interventions that reduced customer outflows by 15% within three months of launch.
- Pioneered flow-of-funds analytics for FHSA and deposit products, decomposing customer lifecycle behaviour — acquisition, attrition, and new-money growth — to inform go-to-market and product strategy.
- Led development of a SARIMAX-based volume forecasting framework supporting balance-sheet planning and revenue/capital allocation decisions, achieving 93% six-month forecast accuracy.
- Envisioned an iterative time-series forecasting framework applied to credit card lifetime value and credit-loss (PD, bankruptcy) forecasting, directly informing P&L and revenue planning.
- Coordinated and led a cross-functional team to design a multi-agent GenAI PoC that automates customer-primacy insights, increasing team leverage and accelerating deposit growth initiatives.
- Led the team's migration to cloud-native GCP pipelines (BQ, Composer, Workstation), building scalable, self-serve analytics infrastructure that increased reliability and team leverage.
- Mentored and coached a team of data scientists to deliver six predictive models spanning attrition, pricing, and growth, generating \$1 million CAD in retention benefit per wave while building team capability.

Senior Data Scientist

Oct 2022 – July 2023

- Partnered with Sales and business transformation teams to deploy and measure ROI for salesforce customer-prioritization, generating \$7MM in annual revenue through model-driven execution and adoption.
- Designed and developed ML-based propensity and response models for Marketing and client engagement, generating \$5.2 million in annual benefits validated through rigorous lift and experimentation frameworks.

- Coached and developed a team of data scientists on best practices in model development, selection, validation, and documentation, raising the analytical bar across the group.

Data Scientist

Mar 2020 – Oct 2022

- Partnered with Product and Operations to build data strategy and productionize self-serve dashboards, KPI definitions, and decision tools supporting go-to-market product decisions.
- Led a multi-business analytics initiative driven by customer intent signals, generating \$400K revenue within six months and transitioning ownership through coaching and repeatable analytical assets.
- Built automated ML monitoring and early-warning indicators unlocking \$112K in annual savings.

University of Washington

Seattle, USA

Research Associate, Finance and Business Economics

Sep 2014 – Jun 2019

Collaborated with professors in documentation, traditional and alternative data management (large and complex) and performing quantitative analysis to address research questions within finance.

- Scraped and analyzed over 18 million global news articles using C#, Python, and GCP NLP APIs for large-scale sentiment analysis.
- Replicated major equity-pricing puzzles using econometric modeling to identify underlying common risk factors.

OTHER ROLES

Frontline Capital Analysts Pvt. Ltd.

Bangalore, India

Associate Director, Credit Risk Analysis

Feb 2014 – May 2014

Infosys Ltd.

Chennai, India

Systems Engineer, Financial Services Industry

Aug 2009 – Dec 2011

EDUCATION

University of Washington – Michael G. Foster School of Business

Seattle, USA

Ph.D., Finance and Business Economics

Class of 2019

- Explained equity returns to default risk firms through multi-factor regression, logistic default prediction, and cash-flow forecasting models.

Cranfield University – Cranfield School of Management

Cranfield, UK

M.Sc. in Finance and Management (Honors)

Class of 2013

- Demonstrated empirical research skills through a master thesis linking corporate default risk with profitability and leverage, employing Datastream, FactSet, MS Excel and VBA.

Amrita Vishwa Vidyapeetham

Coimbatore, India

Bachelor of Technology, Computer Science and Engineering

Class of 2009

- Developed a semantic search engine for smart-home entertainment using clustering and classification algorithms.

LEADERSHIP, COMMUNITY, & LEARNING

- Co-led a division of Women in Data & Analytics within Scotiabank and led enterprise-wide speaker series connecting macroeconomic themes to customer behaviour and analytics strategy.
- Coached students in University of Waterloo's DataFest 2022 on exploratory analysis, model development, and narrative storytelling.
- Committed to upskilling through external courses to strengthen technical skills and business acumen.